VALUATION BOUNDS IN A QUASI-FINITE EXTENSION OF A ONE-DIMENSIONAL LOCAL DOMAIN, WITH EQUALITY EXACTLY IN THE FINITE CASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof refer to same-chapter statements and to the definition of the order function; the latter is recorded with kind definition in the navigation stubs, and the arguments consuming them stay inside the counted span.

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Allowed standard prerequisites (editorial).

- Finiteness detected on the integral closure. Invoked in the counted proof (lines 31205-31209). Replaces the extension by a finite subalgebra of the integral closure, which is where the comparison of orders can be made.
- The order function of a one-dimensional local domain. Used in the inequality and in the equality clause (lines 31193-31201). Supplies the lengths appearing on both sides of the displayed estimate.

Source reference. Original definition 02MD: sources/stacks-algebra/pinned-original-source.tex, source lines 30020–30020. Navigation pointer for \ref{definition-ord} used inside the counted span (line 31200); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02MJ: sources/stacks-algebra/pinned-original-source.tex, source lines 30255–30255. Navigation pointer for \ref{lemma-finite-extension-dim-1} used inside the counted span (line 31215); the argument that consumes it is retained in the span.

Source reference. Original lemma 02MA: sources/stacks-algebra/pinned-original-source.tex, source lines 27820–27820. Navigation pointer for \ref{lemma-finite-in-codim-1} used inside the counted span (line 31205); the argument that consumes it is retained in the span.

Source reference. Original lemma 00QB: sources/stacks-algebra/pinned-original-source.tex, source lines 31123–31123. Navigation pointer for \ref{lemma-quasi-finite-open-integral-closure} used inside the counted span (line 31207); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $A \subset B$ be an extension of domains. Assume*

(1) *A is a local Noetherian ring of dimension 1,*
(2) *$A \to B$ is of finite type, and*
(3) *the induced extension L/K of fraction fields is finite.*

Then B is semi-local. Let $x \in \mathfrak{m}_A$, $x \neq 0$. Let $\mathfrak{m}_i$, $i = 1, \ldots, n$ be the maximal ideals of B. Then

$$[L : K] ord_A(x) \geq \sum\nolimits_i [\kappa(\mathfrak{m}_i) : \kappa(\mathfrak{m}_A)] ord_{B_{\mathfrak{m}_i}}(x)$$

where ord is defined as in Definition 02MD. We have equality if and only if $A \to B$ is finite.

Proof. The ring B is semi-local by Lemma 02MA. Let B' be the integral closure of A in B. By Lemma 00QB we can find a finite A-subalgebra $C \subset B'$ such that on setting $\mathfrak{n}_i = C \cap \mathfrak{m}_i$ we have $C_{\mathfrak{n}_i} \cong B_{\mathfrak{m}_i}$ and the primes $\mathfrak{n}_1, \ldots, \mathfrak{n}_n$ are pairwise distinct. The ring C is semi-local by Lemma 02MA. Let $\mathfrak{p}_j$, $j = 1, \ldots, m$ be the other maximal ideals of C (the "missing points"). By Lemma 02MJ we have

$$\mathrm{ord}_A(x^{[L:K]}) = \sum\nolimits_i [\kappa(\mathfrak{n}_i) : \kappa(\mathfrak{m}_A)] \mathrm{ord}_{C_{\mathfrak{n}_i}}(x) + \sum\nolimits_j [\kappa(\mathfrak{p}_j) : \kappa(\mathfrak{m}_A)] \mathrm{ord}_{C_{\mathfrak{p}_j}}(x)$$

hence the inequality follows. In case of equality we conclude that $m = 0$ (no "missing points"). Hence $C \subset B$ is an inclusion of semi-local rings inducing a bijection on maximal ideals and an isomorphism on all localizations at maximal ideals. So if $b \in B$, then $I = \{x \in C \mid xb \in C\}$ is an ideal of C which is not contained in any of the maximal ideals of C, and hence $I = C$, hence $b \in C$. Thus $B = C$ and B is finite over A. $\square$